

H17 · Behavior MSM with metastable sets; t_2^* as a per-period order parameter (exploratory round 1, 2026-10-03)

States: H14 coarse 6-state minute grid (action classes); pooled non-reversible MSM per goal period, $\tau_c = 5$ min; 27 non-holdout periods.

- Slowest process = tool mode: GUI work (browse+type) vs. shell/chat/idle in most periods (idle alone only in #37, #51).
- $t_2^* = 8.8$ min (median, regime III) vs 5.1 min (regime I): regime III is SLOWER (predicted the reverse).
- Sets beyond sticky states (P3b, Holm): 3/8 regime-III periods; the sojourn null reproduces $\sim 80\%$ of t_2^* .
- Not Markov at minute scale: ITS rise $\times 5.4$ from $\tau = 1$ to 15 min; CK fails in all 27 periods.
- t_2^* does not track error share or output (within-period meta $\rho = -0.18$); agents differ ($I^2 \geq 0.5$ in 96%).
- Verdicts: {'failed': 14, 'mixed': 13}. Jev-like noisy labels would bias t_2 low 4–14 $\times$ unless the shifted estimator is used.

